

SYLLABUS :-

Rock Mechanics: x Physico-mechanical properties of rocks; elastic, plastic and time dependent behaviour of rock. x Stress-strain analysis and relationships x Rock mass classification, engineering properties of rock joints, RMR and Q. x In-situ stress and strength measurement. x Stress distribution around underground openings and design for single opening x Natural and Artificial support systems and their design principles for mine excavation. x Theories of rock failure and applications. x Rock slope and dump stability analysis and design for planer and circular failure. x Concept of mine subsidence and prediction.

3-1-0 Text Books: 1. D. Deb and A. K. Verma, 2016 Fundamentals and Applications of Rock Mechanics, PHI Publications, Chapter - 1 to 9, 11 - 13. 2. W. G. Pariseau 2017, Design Analysis in Rock Mechanics, 3rd Edition, CRC Press, Chapter 2 - 6, 8. 3. John A. Hudson, John P. Harrison 1997 Engineering Rock Mechanics - An Introduction to the Principles, Elsevier Science, Chapter 3 - 8, 11 - 12, 15 - 20. 4. John A. Hudson, John P. Harrison 2000 Engineering Rock Mechanics Part II - Illustrative Worked Examples, Elsevier Science, Chapter 3 - 8, 11 - 12, 15 - 20. 5. B.H.G. Brady, E.T. Brown 2004 Rock Mechanics for Underground Mining. 3rd Edition, Chapman and Hall, London, Chapter 2, 4 - 5, 7-8, 11, 13 - 14. 6. R. E. Goodman 1988 Introduction to Rock Mechanics, 2nd Edition, Wiley, Chapter 2 - 4, 7 - 8. 7. L. Obert, W. I. Duvall 1967 Rock Mechanics and the Design of Structures, John Wiley and Sons Inc.